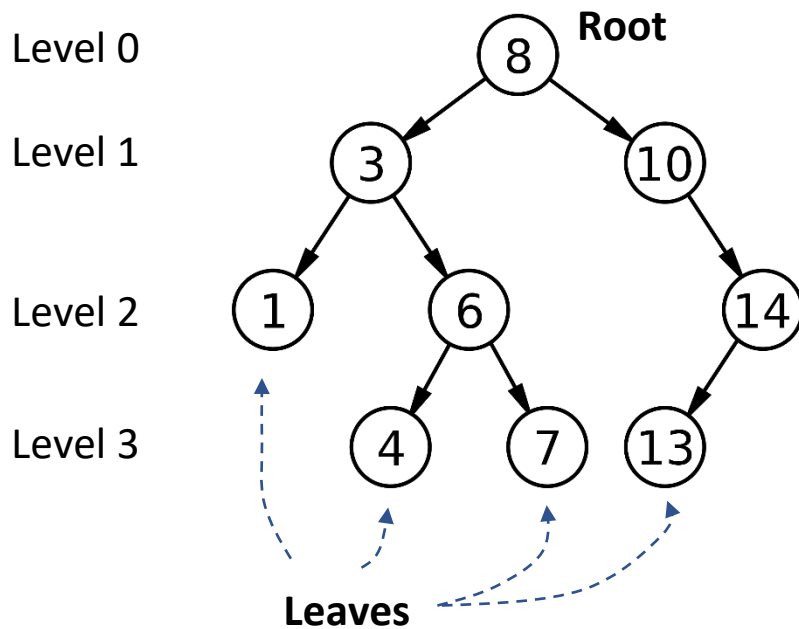


Binary Search Tree Terms

A **binary tree** is a tree data structure in which each node has at most two children, which are referred to as the left child and the right child.

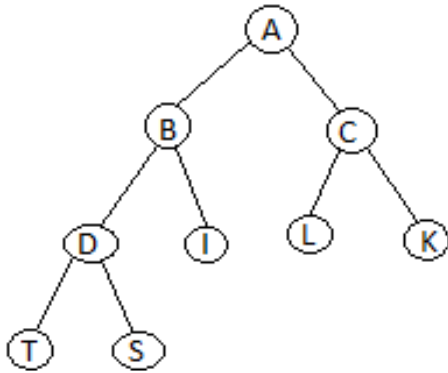


root	the node from which all other nodes "descend".
leaf	a node with no children.
parent	A node which has children. Also known as an internal node.
child	A node which has a parent. The only node that isn't a child is the root.
subtree	A node n and all its descendants. Node n is called the root of the subtree.
depth of a node	the number of edges from the node to the root.
height of a node	the number of edges from node to the lowest leaf in the subtree that has this node as its root.
height of the tree	the length of the path from the root to the bottom node.
diameter	the number of nodes on the longest path between any two nodes in the tree. • Does necessarily go through the root! • You're counting <u>nodes</u>, not links

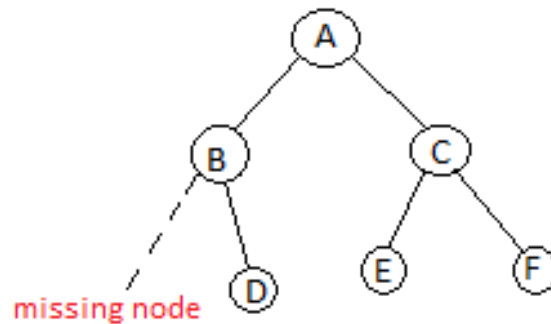
Special Forms of Binary Trees

Complete: In a complete tree, every level above the bottom level is filled.
The bottom level has all nodes shifted as far left as possible.

Complete

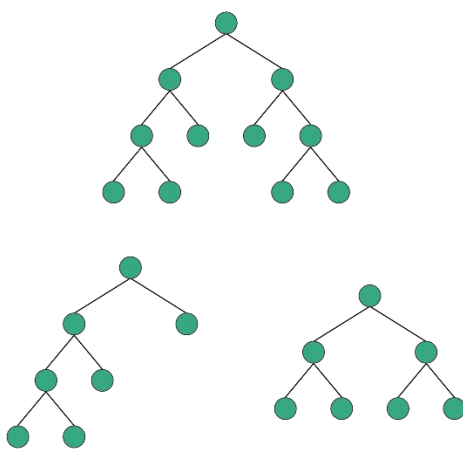


In-complete

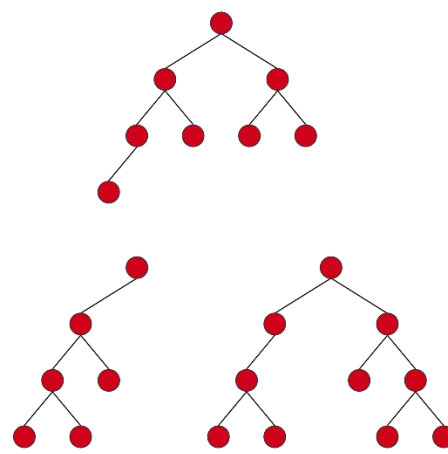


Full / Proper Binary Tree: Strictly binary. Every node has two children or none at all.

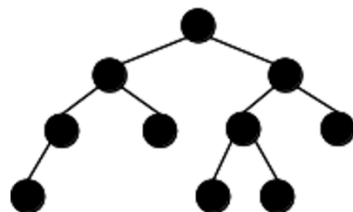
Full / Proper



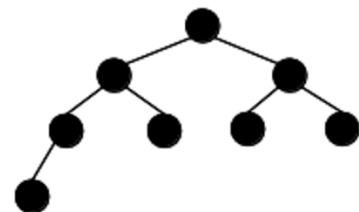
Not full / Improper



Neither complete nor full

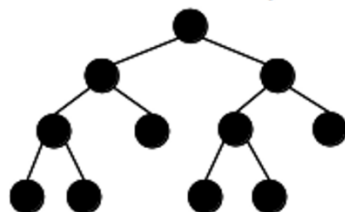


Complete but not full

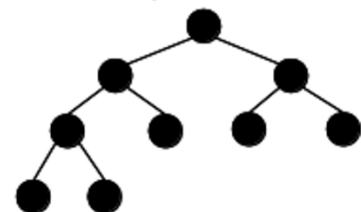


Complete vs. Full trees

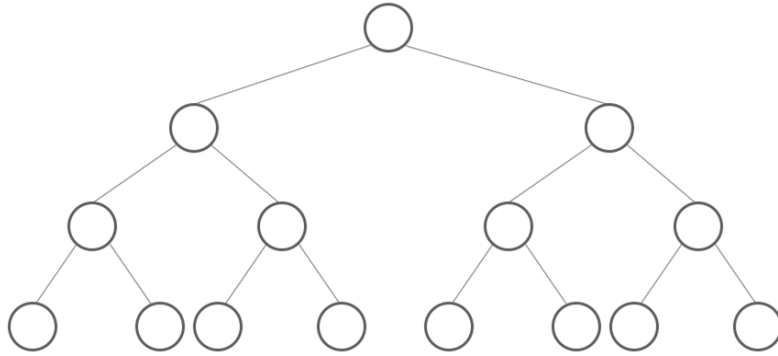
Full but not complete



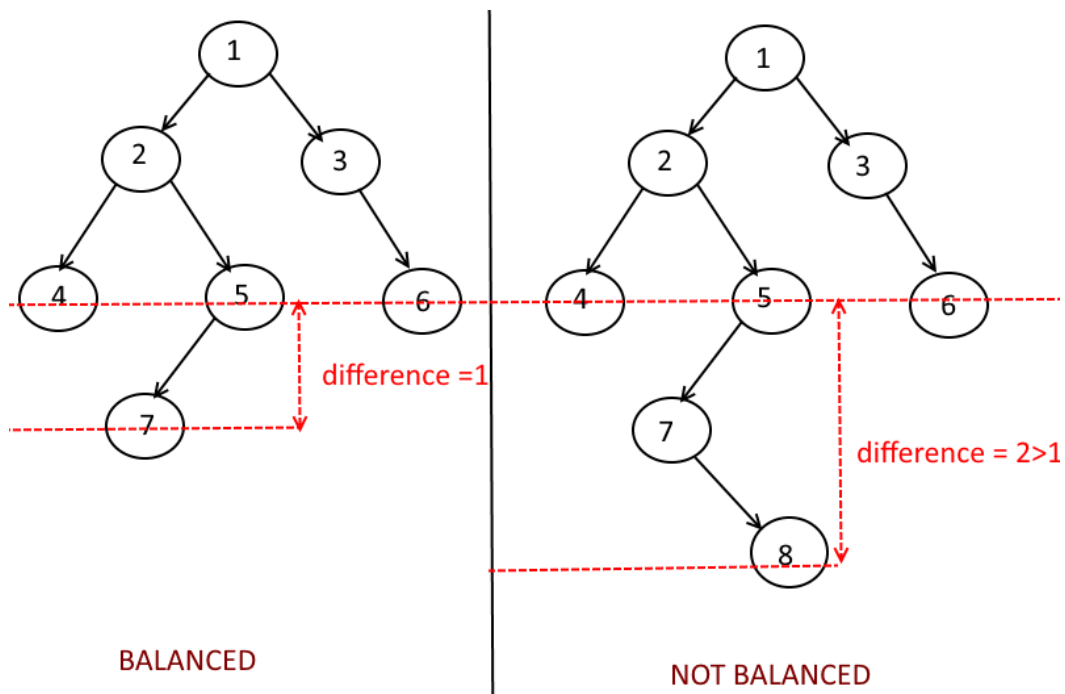
Both complete and full



Perfect: A binary tree is a complete binary tree with the last level also filled completely.



A **balanced binary tree** is a binary tree structure in which the left and right subtrees of every node differ in height by no more than 1.



Degenerate: A tree where each parent node has only one associated child node. This means that performance-wise, the tree will behave like a linked list.

